

VEDA TECHNOLOGY

Data Analytics Internship

Basic SQL SELECT

Task 21 Report

Querying the Northwind database using SELECT, WHERE and ORDER BY

Submitted by

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Data Analytics Track | Level 1 | Day 21

1. Introduction

Structured Query Language (SQL) is the standard language for working with data stored in relational databases. This task builds the fundamentals of SQL by writing simple queries with the **SELECT**, **WHERE** and **ORDER BY** clauses on the Northwind sample database, which represents a small trading company with customers, products, orders and employees.

2. Objective

- Build SQL fundamentals by retrieving data from real business tables.
- Filter rows using WHERE conditions and the IN operator.
- Sort results using ORDER BY and limit output using LIMIT.
- Write 10 clean, well-formatted queries and record their outputs.

3. Tools and Environment

Tool	Purpose
Google Colab	Notebook environment used to write and run the queries
SQLite	Database engine (Excel sheets loaded as tables)
Python (pandas)	Read the Excel file and display query results as tables
Northwind dataset	Sample business database used for all queries

4. Dataset Overview

The Northwind database contains information about a food trading company. The tables used in this task are listed below.

Table	Rows	Description
customers	91	Companies that buy products
products	77	Items sold, with price and stock
orders	830	Orders placed by customers
employees	9	Staff who handle the orders
order_details	2155	Products included in each order
categories	8	Product categories

5. Queries and Outputs

Each query below is shown with its purpose, the SQL code, the total number of rows returned, and a preview of the output.

Q1 Retrieve all customers

```
SELECT *  
FROM customers;
```

91 rows returned (11 columns, first 5 shown) - first 5 rows previewed

customer_id	company_name	contact_name	contact_title	address
ALFKI	Alfreds Futterkiste	Maria Anders	Sales Representative	Obere Str. 57
ANATR	Ana Trujillo Emparedados y helados	Ana Trujillo	Owner	Avda. de la Constitución 2222
ANTON	Antonio Moreno Taquería	Antonio Moreno	Owner	Mataderos 2312
AROUT	Around the Horn	Thomas Hardy	Sales Representative	120 Hanover Sq.
BERGS	Berglunds snabbköp	Christina Berglund	Order Administrator	Berguvsvägen 8

Q2 Select specific columns (company name and country)

```
SELECT company_name, country  
FROM customers;
```

91 rows returned - first 5 rows previewed

company_name	country
Alfreds Futterkiste	Germany
Ana Trujillo Emparedados y helados	Mexico
Antonio Moreno Taquería	Mexico
Around the Horn	UK
Berglunds snabbköp	Sweden

Q3 Customers from Germany

```
SELECT *
FROM customers
WHERE country = 'Germany';
```

11 rows returned (11 columns, first 5 shown) - first 5 rows previewed

customer_id	company_name	contact_name	contact_title	address
ALFKI	Alfreds Futterkiste	Maria Anders	Sales Representative	Obere Str. 57
BLAUS	Blauer See Delikatessen	Hanna Moos	Sales Representative	Forsterstr. 57
DRACD	Drachenblut Delikatessen	Sven Ottlieb	Order Administrator	Walserweg 21
FRANK	Frankenversand	Peter Franken	Marketing Manager	Berliner Platz 43
KOENE	Königlich Essen	Philip Cramer	Sales Associate	Maubelstr. 90

Q4 Products with price greater than 50

```
SELECT product_name, unit_price
FROM products
WHERE unit_price > 50;
```

7 rows returned - first 5 rows previewed

product_name	unit_price
Mishi Kobe Niku	97.0
Carnarvon Tigers	62.5
Sir Rodney's Marmalade	81.0
Thüringer Rostbratwurst	123.79
Côte de Blaye	263.5

Q5 Products sorted by price (high to low)

```
SELECT product_name, unit_price
FROM products
ORDER BY unit_price DESC;
```

77 rows returned - first 5 rows previewed

product_name	unit_price
Côte de Blaye	263.5
Thüringer Rostbratwurst	123.79
Mishi Kobe Niku	97.0
Sir Rodney's Marmalade	81.0
Carnarvon Tigers	62.5

Q6 Customers sorted alphabetically (A-Z)

```
SELECT company_name, city
FROM customers
ORDER BY company_name ASC;
```

91 rows returned - first 5 rows previewed

company_name	city
Alfreds Futterkiste	Berlin
Ana Trujillo Emparedados y helados	México D.F.
Antonio Moreno Taquería	México D.F.
Around the Horn	London
B's Beverages	London

Q7 Orders placed in 1997

```
SELECT order_id, order_date
FROM orders
WHERE strftime('%Y', order_date) = '1997';
```

408 rows returned - first 5 rows previewed

order_id	order_date
10400	1997-01-01
10401	1997-01-01
10402	1997-01-02
10403	1997-01-03
10404	1997-01-03

Q8 Top 5 most expensive products

```
SELECT product_name, unit_price
FROM products
ORDER BY unit_price DESC
LIMIT 5;
```

5 rows returned

product_name	unit_price
Côte de Blaye	263.5
Thüringer Rostbratwurst	123.79
Mishi Kobe Niku	97.0
Sir Rodney's Marmalade	81.0
Carnarvon Tigers	62.5

Q9 Customers from the USA or UK

```
SELECT company_name, country
FROM customers
WHERE country IN ('USA', 'UK');
```

20 rows returned - first 5 rows previewed

company_name	country
Around the Horn	UK
B's Beverages	UK
Consolidated Holdings	UK
Eastern Connection	UK
Great Lakes Food Market	USA

Q10 Employees based in London

```
SELECT first_name, last_name, city
FROM employees
WHERE city = 'London';
```

4 rows returned

first_name	last_name	city
Steven	Buchanan	London
Michael	Suyama	London
Robert	King	London
Anne	Dodsworth	London

6. Conclusion

In this task, I wrote and ran 10 basic SQL queries on the Northwind dataset. I loaded the dataset from Excel into Google Colab, converted it into a SQLite database, and executed the queries.

What I did

- Used **SELECT** to retrieve data from the customers, products and employees tables.
- Used **WHERE** to filter data by country, price and city, including the **IN** operator.
- Used **ORDER BY** to sort results by price (high to low) and company name (A-Z).
- Used **LIMIT** to fetch the top 5 most expensive products.

What I learned

- How to select, filter and sort data using SQL.
- How to write clean, readable queries with capitalised keywords and one clause per line.
- How to work with real business data such as customers, orders and products.

This task built a solid foundation in SQL fundamentals, which will be useful for learning joins, aggregations and data analysis in the upcoming tasks.

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